

Uniqueness of the Rank-5 Projection in the Quantum Gauge Theory of Time

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Abstract

We prove that $r = 5$ is the unique admissible rank within the Majorana–Fibonacci–Heegner projective class of the QGT framework. The result is conditional on four structural inputs: Noetherian balance, second-order Fibonacci autosimilar closure, phasorial realisation of the closure capacity by the admissible catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, and operational metric stability. The proof is a two-sided exclusion. The lower bound does not identify the scalar curvature $|\lambda_+ - \lambda_-|^2 = 5$ with either a spacetime dimension or a Hilbert-space dimension. Rather, this curvature fixes a Fibonacci closure capacity. The rank-five lower bound is obtained only after that capacity is realised by five inequivalent rank-one phasorial sectors. The upper bound follows from the metric threshold $w_4 > \alpha > w_5$. Thus $r = 5$ is not a free truncation parameter inside this admissibility class.

Keywords. Non-invertible projection; rank uniqueness; autosimilar closure; boundary metric; Majorana spinor; finite-rank operator.

The QGT programme studies physical structure generated by a Noetherian balance under restricted accessibility, within the Majorana–Fibonacci–Heegner projective class [1, 20, 19]. In this setting the boundary is represented by a finite-rank non-invertible projection

$$\Pi : \mathcal{H}_{\text{bulk}} \longrightarrow \mathcal{H}_{\text{obs}}, \quad \ker \Pi \neq 0.$$

The admissible finite rank is constrained by two independent mechanisms. First, the Fibonacci closure capacity is realised phasorially by the admissible catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, giving the lower bound $r \geq 5$. Second, operational metric stability gives the upper bound $r \leq 5$ through the bilateral threshold $w_4 > \alpha > w_5$, where α is derived independently from the Pincherle scale increment and Fibonacci curvature [21]. This note proves that these two bounds leave only $r = 5$.

Lemma 1 (Phasorial realisation of the closure capacity). *Let*

$$W := |\lambda_+ - \lambda_-|^2 = 5$$

be the Fibonacci closure capacity of the Majorana–Fibonacci admissibility class. This scalar quantity is not itself a dimension, and it is not the dimension of the pre-geometric domain $M_{\text{pre}}^5 = X_{\text{bdy}} \times S_{\chi}^1(\tau)$. The compact factor $S_{\chi}^1(\tau)$ is an internal phasorial fibre, not an additional observable spacetime direction. The rank-five lower bound is obtained only after the closure capacity is realised by five inequivalent phasorial sectors.

Let

$$\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}.$$

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For $n \in \{2, 3, 7, 11\}$, the imaginary quadratic field $\mathbb{Q}(\sqrt{-n})$ has class-number-one arithmetic in the relevant phasorial channel, so its primitive norm form defines one stable rank-one phasorial sector. The remaining element $n = 5$ is the first lock-in channel where class-one rigidity breaks and the projection pole is met. Thus the admissible catalogue consists of four Heegner-stable channels plus one lock-in channel.

For each $n \in \mathcal{N}_{\text{adm}}$, let $s_n(\tau)$ be the corresponding normalised phasor and let

$$\Pi_n^{(1)} := |s_n\rangle\langle s_n|.$$

The five projectors $\{\Pi_n^{(1)} : n \in \mathcal{N}_{\text{adm}}\}$ define five inequivalent rank-one phasorial sectors. Collecting them as a sector decomposition,

$$\mathcal{V}_{\text{ph}} = \bigoplus_{n \in \mathcal{N}_{\text{adm}}} \mathbb{C} s_n, \quad \dim \mathcal{V}_{\text{ph}} = |\mathcal{N}_{\text{adm}}| = 5,$$

where the direct sum counts inequivalent phasorial channels as distinct rank-one superselection sectors of the boundary representation, not as five arbitrary vectors in a single undifferentiated function space. Therefore any boundary representation that resolves the complete admissible phasorial catalogue must satisfy

$$r \geq \dim \mathcal{V}_{\text{ph}} = |\mathcal{N}_{\text{adm}}| = 5.$$

Remark 2 (Rank is not domain dimension). The equality between the number 5 in the rank and the notation M_{pre}^5 is not dimensional. The domain

$$M_{\text{pre}}^5 = X_{\text{bdy}} \times S_{\chi}^1(\tau)$$

contains a compact internal phasorial fibre over the boundary sector. The projection rank $r = 5$ counts the number of inequivalent rank-one phasorial sectors that the boundary representation must resolve. It is therefore a statement about the observable sector space $\mathcal{H}_{\text{obs}}$, not about the number of spacetime dimensions.

Theorem 3 (Uniqueness of the rank-five projection). *Let $\Pi_r : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{obs}}$ be a finite-rank non-invertible boundary projection in the Majorana–Fibonacci–Heegner admissibility class, satisfying:*

- (I) *Noetherian balance under restricted accessibility;*
- (II) *second-order Fibonacci autosimilar closure $v_{k+2} = v_{k+1} + v_k$;*
- (III) *phasorial realisation of the closure capacity by the admissible catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$;*
- (IV) *operational stability of the induced boundary metric.*

Then $r = 5$ is the unique admissible rank.

Proof. Lower bound. By Lemma 1, the scalar curvature

$$|\lambda_+ - \lambda_-|^2 = 5$$

is used only as the closure capacity of the admissibility class. The boundary lower bound is supplied by the phasorial realisation of that capacity: the admissible catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ consists of five inequivalent rank-one phasorial sectors. These sectors are counted in the observable representation space; they are not coordinates of M_{pre}^5 . A boundary projection resolving the complete catalogue must therefore have

$$r \geq \dim \mathcal{V}_{\text{ph}} = |\mathcal{N}_{\text{adm}}| = 5.$$

For $r < 5$, at least one admissible phasorial sector has no representative in the retained boundary space. The Noetherian balance and the second-order autosimilar closure cannot then be represented on the full admissible catalogue. Hence $r \geq 5$.

Upper bound. In the SVD representative of the Fibonacci-autosimilar projection, the singular values are $\sigma_k = \varphi^{-k}$, and the transition weights are

$$w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}.$$

The threshold α is derived independently from the Pincherle scale increment and Fibonacci curvature, not from the rank choice. The bilateral inequality

$$w_4 > \alpha > w_5$$

then separates resolvable from sub-threshold transition cells. If $r > 5$, the retained space includes the transition $k = 5$, whose weight satisfies $w_5 < \alpha$. The induced boundary metric would assign positive distinguishability to a direction below the operational resolution threshold, violating metric stability. Hence $r \leq 5$.

Combining the two bounds gives $r = 5$. □

Corollary 4. *Under the hypotheses of Theorem 3, in the SVD representative of [21]: the retained singular values are $\sigma_k = \varphi^{-k}$, $k = 0, \dots, 4$; the boundary metric is $d_{\partial}^2 = \sum_{k=0}^4 \varphi^{-2k} |c_k|^2$; the threshold α satisfies $w_4 > \alpha > w_5$; in the QGT angular readout convention, $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$.*

Scope. The theorem is conditional on the Majorana–Fibonacci–Heegner admissibility class. It does not re-derive the Majorana condition, the Fibonacci autosimilar structure, or the phasorial catalogue from the bare conservation law alone. Its result is narrower: once those structural inputs are fixed, the finite-rank boundary projection cannot choose its rank freely. The complete admissible catalogue requires at least five rank-one phasorial sectors, while metric stability excludes all retained directions beyond the fifth. Thus $r = 5$ is unique inside the class.

Downstream QGT readouts that use $r = 5$ remain subject to their own declared hypotheses and open gates.

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